

VOSSEVANGEN, Norway

WMO: 013370

Lat: 60.6250N

Lon: 6.4261E

Elev: 180

StdP: 14.60

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.5	(c) 3.4	(d) -9.1	(e) 3.4	(f) -4.8	(g) -1.0	(h) 5.2	(i) 4.0	(j) 13.6	(k) 37.4	(l) 12.1	(m) 36.3	(n) 0.8	(o) 60	(p) 0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	80.1	62.5	76.0	60.6	72.3	58.7	64.2	75.5	62.4	73.0	60.6	69.3	4.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.7	77.1	67.2	58.1	72.6	65.6	56.5	68.5	64.3	29.4	75.3	28.1	73.3	26.9	69.3	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.0	(b) 9.0	(c) 7.7	(e) -2.3	(f) 86.0	(g) 9.4	(h) 4.1	(i) -9.1	(j) 88.9	(k) -14.6	(l) 91.3	(m) -19.8	(n) 93.6	(o) -26.7	(p) 96.6		
(a) 10.0	(b) 9.0	(c) 7.7	(e) -2.9	(f) 66.6	(g) 9.3	(h) 3.2	(i) -9.5	(j) 68.9	(k) -15.0	(l) 70.8	(m) -20.2	(n) 72.6	(o) -26.9	(p) 74.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	27.0	27.9	34.3	42.1	50.1	56.6	61.8	59.1	52.5	43.3	35.3	29.6		
	DBStd	14.16	11.34	10.32	7.21	5.33	6.37	5.01	5.71	4.33	5.25	6.35	7.79	10.59		
	HDD50	3475	713	617	488	244	78	6	0	0	33	221	441	634		
	HDD65	7935	1178	1037	953	687	463	257	135	189	374	672	891	1099		
	CDD50	1062	0	0	0	7	82	202	366	282	109	14	0	0		
	CDD65	47	0	0	0	0	1	4	36	6	0	0	0	0		
	CDH74	551	0	0	0	0	41	106	348	55	1	0	0	0		
	CDH80	110	0	0	0	0	8	15	83	4	0	0	0	0		
(14) Wind	WSAvg	3.3	3.6	3.3	3.6	3.5	3.7	3.5	3.2	3.0	3.0	3.3	3.3	3.2		
(15) Precipitation	PrecAvg	86.1	9.7	8.6	8.1	4.4	3.6	4.3	4.8	5.6	7.5	9.3	8.6	10.0		
	PrecMax	122.6	19.3	18.4	23.9	8.3	8.0	9.0	7.8	9.0	13.5	21.1	17.3	19.2		
	PrecMin	51.3	0.2	0.1	1.0	1.4	0.6	1.2	1.5	2.3	2.2	2.4	1.1	1.2		
	PrecStd	15.0	5.4	5.1	4.9	1.8	1.9	2.0	1.4	2.0	3.4	4.4	4.3	5.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.6	46.8	53.8	66.5	78.7	82.2	87.1	79.8	71.8	64.4	52.3	48.1		
		MCWB	39.8	39.8	43.3	51.2	59.6	60.9	66.1	63.0	58.8	53.6	47.5	42.7		
	2%	DB	42.2	43.7	48.2	61.3	73.3	77.0	81.9	74.8	67.1	58.4	49.6	44.7		
		MCWB	37.9	38.0	41.1	48.3	56.3	58.6	64.1	60.8	56.9	51.4	45.5	40.4		
	5%	DB	39.8	41.1	45.0	57.0	68.9	72.5	77.9	71.2	63.6	54.4	47.3	42.3		
		MCWB	36.1	36.4	39.4	45.9	53.9	56.5	62.4	59.3	55.4	49.5	44.0	38.6		
	10%	DB	37.8	38.9	42.8	53.0	63.8	68.5	73.6	67.8	60.7	52.1	44.7	40.3		
		MCWB	34.8	35.5	38.2	43.8	51.5	55.0	60.8	58.3	54.4	47.8	42.1	37.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.9	42.1	45.0	52.6	60.4	62.7	67.8	64.7	61.1	55.1	48.5	44.0		
		MCDB	44.0	45.6	51.0	62.9	75.3	77.6	82.9	76.4	68.5	60.8	50.8	46.8		
	2%	WB	38.5	39.2	42.5	49.3	57.6	60.3	65.5	62.6	58.6	52.5	46.5	41.2		
		MCDB	41.3	42.5	46.9	59.0	71.1	73.7	78.3	71.7	63.9	56.9	49.0	43.4		
	5%	WB	36.9	37.3	40.3	47.1	54.8	58.2	63.7	60.9	56.8	50.4	44.3	39.4		
		MCDB	39.2	40.1	44.2	55.6	66.6	68.3	74.6	68.3	61.8	54.2	46.5	41.6		
	10%	WB	35.4	35.8	38.7	44.7	52.3	56.4	62.0	59.3	55.1	48.3	42.1	37.8		
		MCDB	37.2	38.3	42.2	51.4	62.6	65.3	72.1	65.5	59.9	51.3	44.3	39.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	10.0	13.1	17.5	19.0	18.5	17.0	15.2	13.3	11.0	7.7	7.9		
		MCDBR	9.5	9.5	15.0	24.9	28.9	28.5	26.1	22.3	18.0	13.2	10.3	9.3		
	5% WB	MCWBR	6.7	6.0	9.4	13.8	14.0	12.9	10.6	10.5	9.8	8.4	7.3	6.8		
		MCWBR	8.5	9.1	13.7	22.3	26.8	23.1	23.5	18.3	15.9	11.7	9.3	8.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.257	0.297	0.333	0.371	0.373	0.367	0.381	0.374	0.345	0.323	0.273	0.232		
		taud	2.083	2.170	2.142	2.122	2.296	2.387	2.377	2.392	2.460	2.399	2.173	2.002		
	Ebn at Noon	Ebn at Noon	151	216	246	258	269	273	266	258	246	210	145	106		
		Edh at Noon	13	23	32	40	36	34	34	31	24	18	12	9		
(44) All-Sky Solar Radiation	RadAvg	78	241	639	1215	1523	1528	1456	1132	713	330	109	51			
	RadStd	19	67	123	201	249	180	175	104	85	48	17	7			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page